

Muhammad Irfan

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PROFESSIONAL SUMMARY

Motivated 2nd-year Computer Science student with a strong academic record (Dean's List) and hands-on experience across AI/Computer Vision, systems programming, and OOP. Hackathon winner with a proven ability to ship high-performance solutions under competition conditions. Seeking an entry-level software engineering or AI/ML role to apply research-grade skills in a real-world environment.

EDUCATION

FAST – NUCES

Bachelor of Science in Computer Science | CGPA: **3.73 / 4.0**

2024 – Present
Karachi, Pakistan

- **Dean's List:** 2nd and 3rd Semesters — awarded for academic excellence
- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, OOP, Software Design & Analysis, Database Systems, Digital Logic Design, Discrete Mathematics, Computer Organization & Assembly Language

HONORS & ACHIEVEMENTS

Runner Up — Computer Vision AI Hackathon (Duality x AIT): built a Secure Object Detection model achieving the highest mAP (Mean Average Precision) among all competing teams

Image Segmentation Expert — successfully trained SegFormer MiT-B2 transformer model on Kaggle for semantic segmentation tasks

Dean's List Certificate — awarded Semesters 2 and 3 for sustained academic excellence at FAST-NUCES

EXPERIENCE

Research & Development (Self-directed)

2024 – Present

AI/Computer Vision Domain

- Independently studied and applied state-of-the-art Computer Vision techniques including transformer-based segmentation and object detection
- Participated in competitive hackathon environment, delivering a **production-quality model** within a time-constrained setting
- Actively exploring **LLM-OS integration** — researching how large language models can interface with OS kernel architectures (in progress)

PROJECTS

Secure Object Detection Model | Python, YOLO, OpenCV

2025

- Developed an object detection pipeline that achieved the **highest mAP score** in the Duality x AIT Computer Vision Hackathon, placing **2nd overall**
- Optimized inference speed and precision trade-off to maximize evaluation metrics under competition constraints
- Demonstrated low-level systems thinking with direct register and memory manipulation

Flappy Bird Clone | C++, Qt Framework

2024

- Built a high-performance Flappy Bird clone using C++ and the Qt Framework with smooth animation and collision detection
- Implemented game loop, physics simulation, and score tracking using Qt's graphics and event systems
- Available on GitHub: github.com/MuhammadIrfannnnn/Flappy-Bird-Qt-CPP

- Implemented an optimized pathfinding system using graph algorithms (Dijkstra/BFS) for fastest route computation
- Applied advanced DSA principles including priority queues and adjacency list representation

ONGOING / IN PROGRESS

LLM-OS Architecture: researching integration of large language models with OS-level scheduling and process management

Aviation Tracker: database-driven flight disruption analysis tool using SQL and real-world datasets

SKILLS

Languages : C, C++, Python, SQL, x86 Assembly (COAL), HTML/CSS

AI & Vision : OpenCV, SegFormer (HuggingFace), YOLO-based Object Detection, Kaggle Notebooks

Frameworks : Qt Framework (C++), OOP Design Patterns

Tools : Git, GitHub, VS Code, Linux, Postman

Databases : SQL / PostgreSQL (basics), Firebase (basics)

Design & UML : UML (Class, Sequence, Activity, State, Component, Deployment), ECB, 4+1 View Model, Design Patterns

Soft Skills : Problem Solving, Hackathon Execution, Team Collaboration, Technical Documentation

COURSES & SELF-LEARNING

Kaggle: Computer Vision course — object detection, image segmentation, data augmentation

HuggingFace Transformers: hands-on with SegFormer, BERT-family architectures

GitHub Learning Lab: Git workflows, CI/CD fundamentals